MID-SEMESTER RESEARCH PROJECT MILESTONE

Year 3, Semester 1 | BSc Computer Science and BSc Data Science and Analytics

Wednesday, 14 October 2026 | 8:00 AM-4:00 PM | 5 minutes per student

Instructions

Each student must print ONE single-sided A4 research summary and present it to the panelist. No PowerPoint is required. The purpose is to verify the research direction, technical approach, work completed to date, and readiness to produce a working prototype and research poster in approximately six weeks.

Use the printed page to explain your project briefly (about 2-3 minutes), leaving about 2 minutes for questions. Bring evidence of completed work, such as source code, literature notes, designs, datasets or experiment results, in case the panelist requests it.

Your printed A4 page must contain

Section	Required content
Identification	Full name, registration number, programme, project title.
Problem and objective	The specific problem and main research objective.
Existing research and gap	What previous studies/solutions have done, their limitations, and your evidence-based gap; cite 2-3 academic sources.
Proposed contribution	What your project will investigate, improve, test or establish beyond simply implementing existing work.
Full technical pipeline / architecture	An easy-to-read, project-specific diagram showing data/input type and source, processing, proposed algorithms/models or components, outputs and evaluation. Label what is new or under investigation.
Progress to date	Completed activities and verifiable evidence, distinguished from planned work.
Next six weeks	Key remaining activities, intended working prototype and how you will evaluate it.

Research quality requirement

Using existing data, pretrained models, software libraries or frameworks is permitted. However, merely applying an existing model to an existing dataset with standard training and reporting accuracy, or building a routine application without a substantive investigation, is not sufficient research contribution.

Final semester expectation: a research poster, a functioning prototype, and credible evaluation evidence. The sample on the next page illustrates the required level of specificity; do not copy its research topic or pipeline.

SAMPLE: ONE-PAGE STUDENT RESEARCH SUMMARY

ILLUSTRATIVE EXAMPLE ONLY | Adapt the headings and architecture to your own project.

Registration No.: S25… | Student: Sample Student Name | Program: BSc Data Science and Analytics

Title: Evaluating Low-Label Training Strategies for Pneumonia Classification from Chest X-ray Images

Problem and research objective

Medical-image classifiers may be difficult to adapt when only a small number of locally relevant labelled images are available.

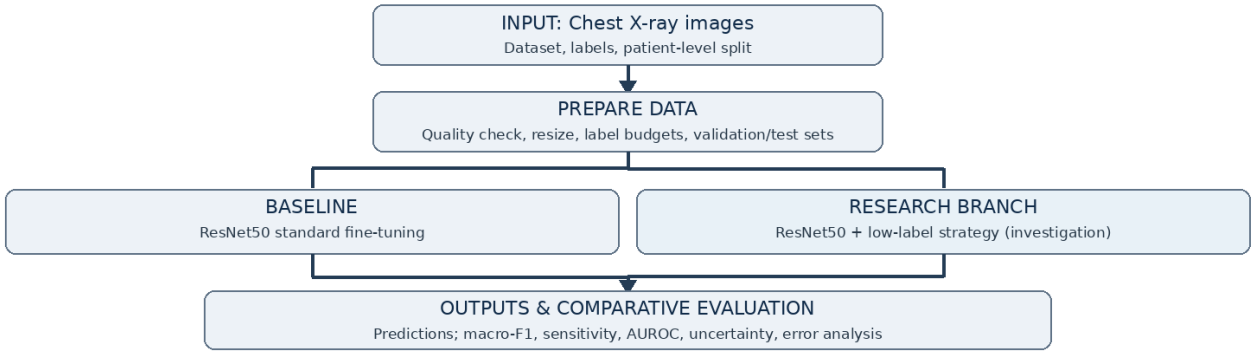
Objective: investigate whether a low-label training strategy improves classification performance and stability compared with standard transfer learning under fixed label budgets.

What exists, gap and proposed contribution

Previous work has investigated convolutional networks, pretrained models and data augmentation for chest X-ray classification. This project will examine a narrower gap: comparative performance when labelled examples are deliberately limited.

Proposed contribution: controlled evidence across several label budgets, with uncertainty and error analysis rather than accuracy alone. Cite and discuss 2-3 directly relevant peer-reviewed studies (replace the prompts below with actual verified references).

Full research pipeline / proposed architecture



Comparison logic: both branches use the same held-out test set and label budgets. Research focus = performance and reliability under limited labels, not simply training ResNet50.

Completed progress (illustrative, replace with genuine evidence)

Literature matrix drafted (10 papers); candidate dataset identified and licence checked; initial project repository and preprocessing script created; baseline experiment planned. No performance claims yet: experiments have not been completed.

Remaining six weeks and expected prototype

W1: confirm dataset and split. W2: run baseline. W3: implement low-label strategy. W4: repeated experiments. W5: evaluate errors and robustness. W6: finalise an inference-and-comparison prototype, poster and results. Prototype: small interface showing sample predictions plus comparative results under selected label budgets.

Key references (replace with actual verified sources)

[1] Author, A. (Year). Paper title. Venue. [2] Author, B. (Year). Paper title. Venue. [3] Author, C. (Year). Paper title. Venue.